

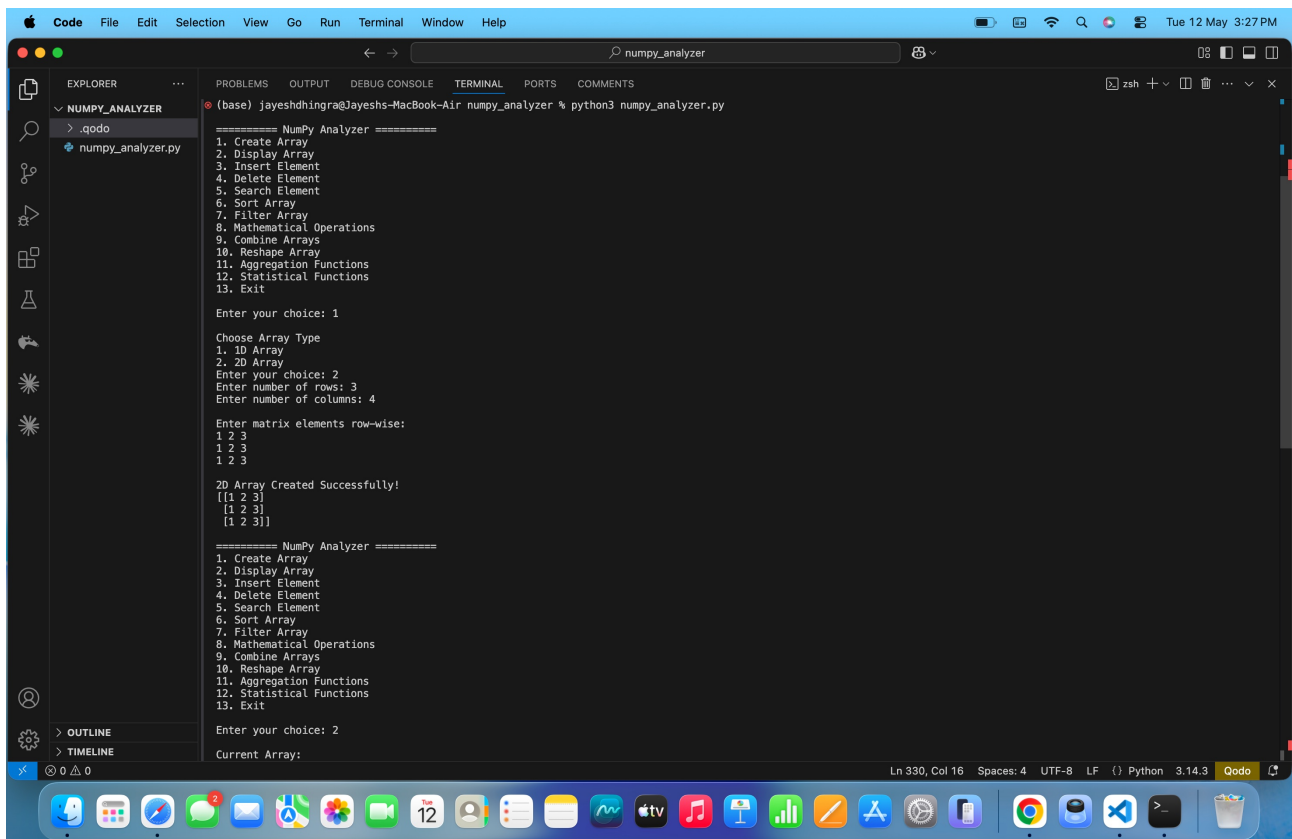
NumPy Analyzer — Output Screenshots

Program Execution Results in VS Code Terminal

RED & WHITE SKILL EDUCATION

The following screenshots capture the live execution of **numpy_analyzer.py** inside the VS Code integrated terminal on macOS. Both screenshots demonstrate the interactive menu-driven console interface working correctly with real user input and NumPy output.

Screenshot 1 — Array Creation (2D) & Search Element



```
(base) jayeshdingra@Jayeshs-MacBook-Air numpy_analyzer % python3 numpy_analyzer.py

===== NumPy Analyzer =====
1. Create Array
2. Display Array
3. Insert Element
4. Delete Element
5. Search Element
6. Sort Array
7. Filter Array
8. Mathematical Operations
9. Combine Arrays
10. Reshape Array
11. Aggregation Functions
12. Statistical Functions
13. Exit

Enter your choice: 1

Choose Array Type
1. 1D Array
2. 2D Array
Enter your choice: 2
Enter number of rows: 3
Enter number of columns: 4

Enter matrix elements row-wise:
1 2 3
1 2 3
1 2 3

2D Array Created Successfully!
[[1 2 3]
 [1 2 3]
 [1 2 3]]

===== NumPy Analyzer =====
1. Create Array
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11. Aggregation Functions
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13. Exit

Enter your choice: 2

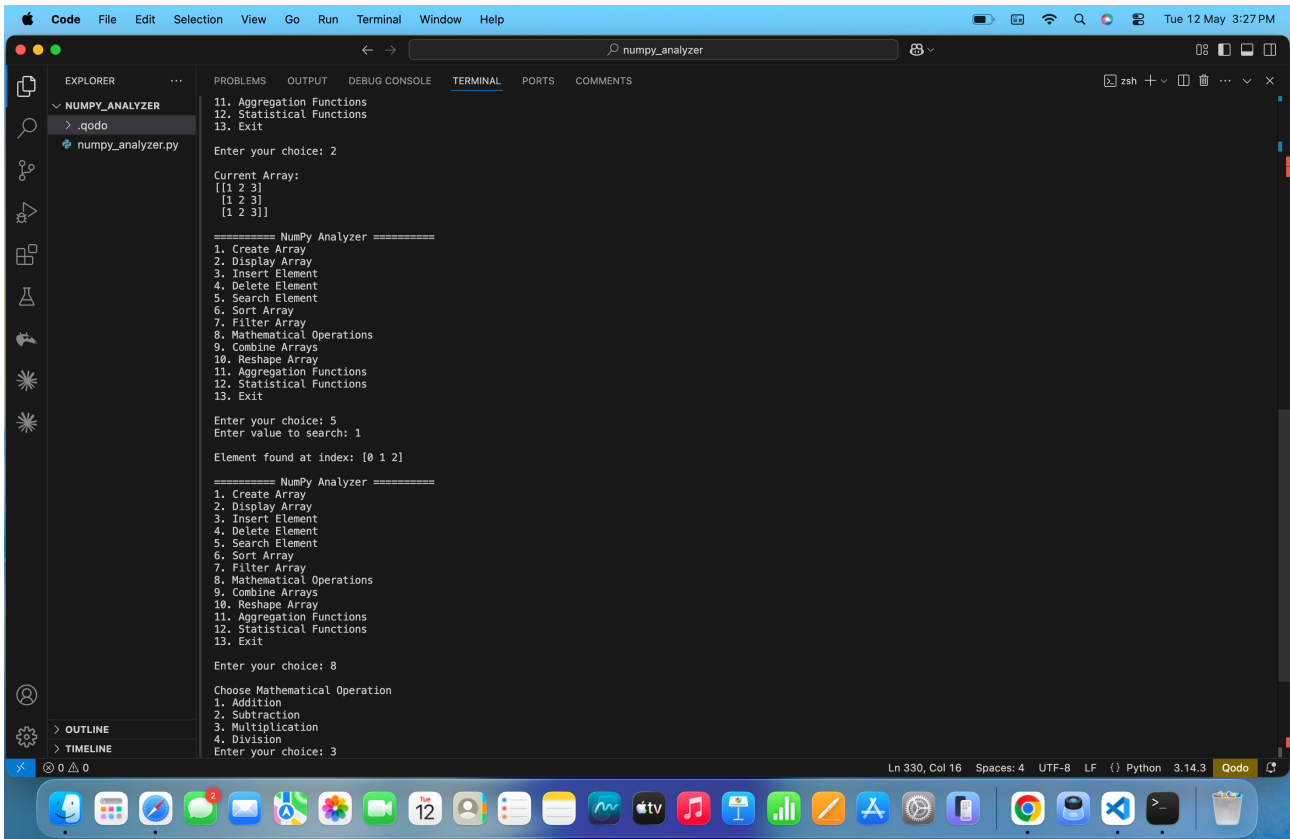
Current Array:
```

Fig 1: User selects option 1 (Create Array) → chooses 2D → enters 3 rows × 4 columns → inputs matrix elements row-wise → 2D array created successfully and displayed. Then selects option 2 (Display Array) to confirm the stored array.

Observation	Detail
Menu displayed	13-option NumPy Analyzer menu printed cleanly
Option chosen	1 → Create Array
Array type	2D Array (3 rows × 4 columns)

Input method	Elements entered row-wise as space-separated values
Output	2D Array Created Successfully! <code>[[1 2 3] [1 2 3] [1 2 3]]</code>
Option 2	Display Array — confirms current stored array

Screenshot 2 — Search Element & Mathematical Operations



```
Code File Edit Selection View Go Run Terminal Window Help
numpy_analyzer

EXPLORER
  NUMPY_ANALYZER
    .qodo
    numpy_analyzer.py

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS
Enter your choice: 2
Current Array:
[[1 2 3]
 [1 2 3]
 [1 2 3]]

===== NumPy Analyzer =====
1. Create Array
2. Display Array
3. Insert Element
4. Delete Element
5. Search Element
6. Sort Array
7. Filter Array
8. Mathematical Operations
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Enter your choice: 5
Enter value to search: 1
Element found at index: [0 1 2]

===== NumPy Analyzer =====
1. Create Array
2. Display Array
3. Insert Element
4. Delete Element
5. Search Element
6. Sort Array
7. Filter Array
8. Mathematical Operations
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11. Aggregation Functions
12. Statistical Functions
13. Exit

Enter your choice: 8
Choose Mathematical Operation
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice: 3
```

Fig 2: User selects option 5 (Search Element) → enters value 1 → program returns 'Element found at index: [0 1 2]'. Then selects option 8 (Mathematical Operations) → chooses Multiplication.

Observation	Detail
Option 5	Search Element — searches for value 1 in the array
Search result	Element found at index: [0 1 2]
Option 8	Mathematical Operations sub-menu displayed
Operations shown	1. Addition 2. Subtraction 3. Multiplication 4. Division
Choice made	3 → Multiplication selected

Execution Summary

Feature	Status	Details
Program launch	✓ Pass	python3 numpy_analyzer.py runs without errors

Menu display	✓ Pass	All 13 options displayed correctly
Array Creation (2D)	✓ Pass	3×4 matrix created and stored successfully
Display Array	✓ Pass	Stored array printed as <code>[[1 2 3][1 2 3][1 2 3]]</code>
Search Element	✓ Pass	Value 1 found at indices <code>[0 1 2]</code>
Math Operations menu	✓ Pass	Sub-menu with 4 operations displayed correctly
Environment	✓ Pass	VS Code terminal, macOS, Python 3.14.3